
A Capable, Fully Tensor-Decomposable Vision-Language-Action Policy

Exact rational conversion, protocol-aligned control, and weight-derived
analysis

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Abstract

1 Tensor decomposition factors a large table of interactions into a small set of
2 low-rank components that can be ranked and inspected. Applying it directly
3 to a conventional robot policy is difficult because softmax, ordinary activations,
4 and normalization interrupt the required algebraic form. We introduce χ -VLA,
5 a compact vision-language-action policy whose learned map uses only bilinear,
6 linear, and rational operations. Its checkpoint therefore defines an explicit rational
7 tensor program, making interactions among image, instruction, state, and action
8 coordinates accessible from the weights. A runtime operator audit of the 20.1M
9 ViT checkpoint finds no incompatible learned operation, and local reconstructions
10 reproduce its deployed rational and bilinear branches near the float32 precision
11 floor. Three historical aggregate records for the convolutional 19.2M-parameter
12 instantiation report $93.7\% \pm 0.8\%$ across all ten LIBERO-Object tasks under the
13 recorded three-seed protocol. On the three released convolutional checkpoint files
14 named in those historical aggregate records, independent reconstruction covers
15 all 4,608 joint-attention head-input cases with worst relative error 6.36×10^{-16} .
16 In a separate three-checkpoint ViT instantiation, we derive an exact weight-based
17 decomposition of softmax-free attention and reconstruct every attention module
18 with worst relative error 2.56×10^{-7} . In a seed-0 ViT checkpoint, all 36 primary
19 block-head tests favor its rank-128 subspace over equal-rank random controls, with
20 median fidelity ratio 2.44. The evidence establishes strong specialist capability
21 and exact layerwise access to trained attention coefficients. It does not establish
22 broad robotic generalization, robust instruction grounding, or an exact end-to-end
23 decomposition of the full policy.

24 1 Introduction

25 Modern robot policies can work well while remaining difficult to inspect. A checkpoint contains
26 millions of learned numbers, but it does not directly reveal which combinations of image regions,
27 instruction tokens, and robot state produce an action. Analysts therefore usually collect activations
28 on a dataset and fit a separate probe. That can identify correlations, but it leaves open whether the
29 same structure is present in the weights and whether it causes behavior.

30 Tensor decomposition offers a different analysis surface. A tensor is a multidimensional table, and a
31 decomposition rewrites that table as a sum of simpler low-rank factors, much as a matrix factorization
32 finds dominant directions in a matrix. If a network is built from additions, multiplications, and ratios
33 of polynomials, its weights define an explicit table of input interactions. Decomposing that table can

rank the combinations the model is constructed to use. By *fully tensor-decomposable*, we mean that every learned block preserves this algebraic form. This is an operator-closure claim, not a claim that we materialize one compact end-to-end coefficient tensor. Its degree and representation size grow rapidly with depth. For example, one coefficient can connect an image coordinate, an instruction coordinate, and an action coordinate. Factoring the coefficient table ranks high-energy cross-modal interactions directly from the checkpoint without first fitting an activation probe. Those factors are candidate structure, not automatically human-readable concepts or causal explanations. The aim is not merely compression. It is a direct path from parameters to candidate mechanisms that can be tested by intervention [3, 4].

Robot control makes this goal useful and difficult. Visual shortcuts can produce high benchmark success while ignoring the instruction, and small action errors compound in closed loop. Standard softmax, nonlinear activations, and square-root normalization break the explicit polynomial form. Continuous actions also cannot rely on the external categorical sampling used by a language model. A practical design must recover these functions without losing precise control.

We study the narrower but testable question:

Can a specialist VLA retain strong closed-loop capability when every deployed learned operation is polynomial or rational, while supporting exact weight analysis?

We test capability and structural analysis in two fully rational encoder instantiations. The convolutional checkpoints establish closed-loop capability. Separate ViT checkpoints support the exact attention audit.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We report three historical aggregate records for a fully rational 19.2M convolutional policy under a protocol aligned with published LIBERO-Object results. The records give $93.7\% \pm 0.8\%$.
3. We mechanically audit the 20.1M ViT checkpoint and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes exact weight structure directly accessible.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive below replaces a familiar transformer component while keeping an explicit coefficient description. Linear maps contribute first-order terms. Multiplying learned linear projections creates higher-order interactions whose coefficients remain functions of the weights, rather than an unstructured nonlinear black box.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

77 The inner sum is an order-three table indexed by output coordinate and two input coordinates. The
 78 three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization,
 79 without materializing every table entry.

80 **Softmax-free bilinear attention.** Each head forms two query-key systems and one value stream:

$$Y = C [M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

81 Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value,
 82 and output maps are linear. Equation 2 is therefore a polynomial in its input activations. Unlike
 83 softmax attention, every multiplicative interaction can be expanded into coefficients determined by
 84 the trained weights.

85 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
 86 root. Let $m(x) = d^{-1} \sum_{j=1}^d x_j^2 + \epsilon$, and let $s_0 > 0$ be the frozen running mean-square scale. We
 87 deploy a fixed rational approximation $r(v) = P(v)/Q(v)$ to $v^{-1/2}$ over $v \in [0.1, 10]$:

$$v = \frac{m(x)}{s_0}, \quad \text{RNorm}(x) = \frac{x}{\sqrt{s_0}} r(v). \quad (3)$$

88 Here *rational* means a ratio of two polynomials. Projective coordinates carry each activation as a
 89 numerator-denominator pair (N, D) representing N/D . Linear, bilinear, residual, and normalization
 90 operations update this pair. The model therefore keeps input-specific rescaling while remaining
 91 exactly representable as a rational tensor network. Training and deployment use r itself. The
 92 exactness claims below therefore concern faithful evaluation of the deployed rational computation,
 93 not uniform agreement with RMSNorm. Appendix A.3 quantifies that distinction.

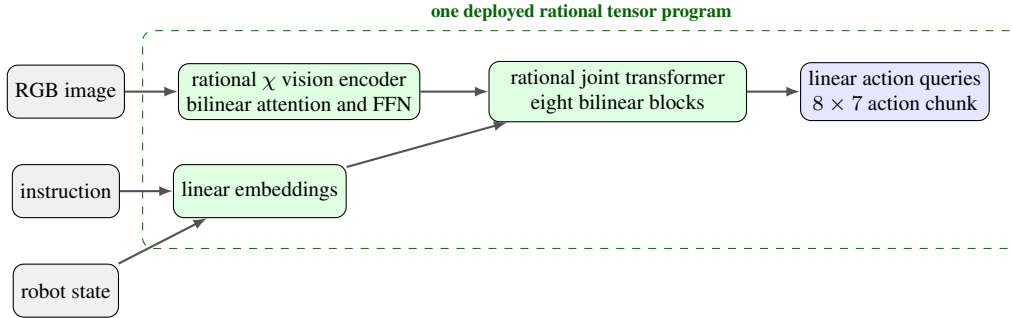


Figure 1: **Deployed χ -VLA map.** The strongest checkpoints use one vision stream and a linear continuous-action head. Every learned component inside the dashed boundary is linear, bilinear, or rational.

94 2.2 Architecture and training

95 Both encoder variants follow the high-level VLA sequence of visual encoding, language and state
 96 embedding, joint processing, and action decoding [1]. They use a 64×64 agent-view image, an
 97 eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. Both share
 98 a width-384, eight-block, twelve-head joint backbone and a linear head that maps each action-query
 99 state to seven continuous action dimensions. The 20.1M ViT uses four bilinear-attention vision
 100 blocks and has 20,137,352 learned parameters. The 19.2M convolutional variant instead uses three
 101 stride-2 bilinear convolution stages that produce an 8×8 token grid. The historical capability records
 102 use the convolutional variant. The runtime operator census and reduced-Gram screen use the ViT. The
 103 released convolutional files additionally receive the joint-attention and rational-normalizer certificates
 104 reported in Appendix A.3.

105 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
 106 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
 107 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
 108 that verify every requested state was loaded.

2.3 Mechanical certificate on the 20.1M ViT checkpoint

Architecture definitions can differ from deployed code. We therefore audit the real 20.1M ViT 189/200 checkpoint at runtime and reconstruct representative operations from saved weights.

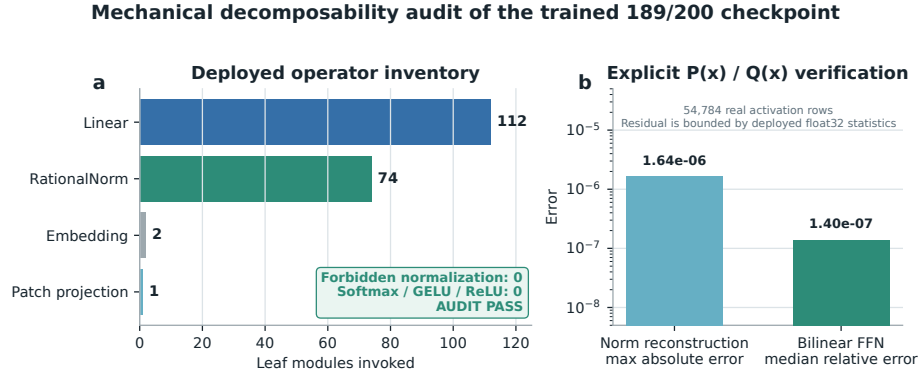


Figure 2: **Mechanical audit of the 20.1M ViT rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.

The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN reconstruction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed module’s float32 statistic. The audit certifies operator membership and local reconstruction. It does not materialize one compact polynomial for the full eight-layer transformer.

3 Protocol-aligned closed-loop capability

3.1 Evaluation protocol

The historical aggregate records specify the following evaluation protocol for all ten LIBERO-Object tasks:

- 50 canonical trials per task
- a 280-step cap
- three independently trained convolutional seeds
- 500 rollouts per seed and 1,500 rollouts in total

Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not rerun in our codebase, so differences may still include implementation and training choices.

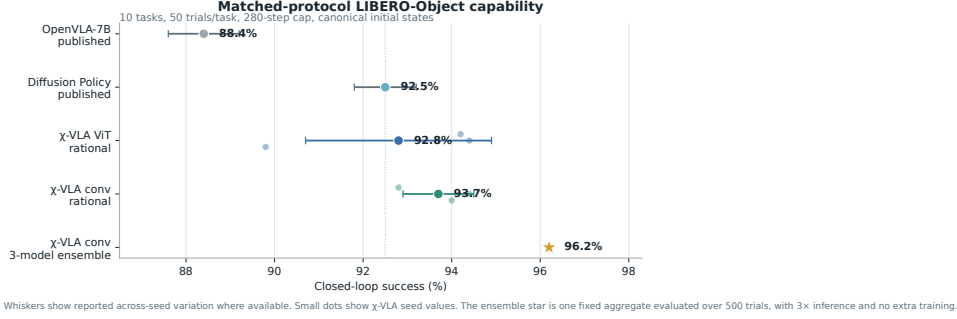


Figure 3: **Protocol-aligned LIBERO-Object capability.** Small dots are historical χ -VLA aggregate seed values. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational conv	19.2M	3	93.7% $\pm$ 0.8%
χ -VLA, conv ensemble	$3 \times 19.2\text{M}$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	88.4% $\pm$ 0.8%
Diffusion Policy [1]	not matched	3	92.5% $\pm$ 0.7%

Table 1: **Closed-loop success under aligned evaluation settings.** The single-model χ -VLA row is the mean $\pm$ sample standard deviation from surviving historical aggregate records. Their original per-episode files are unavailable. The ensemble row comes from a separate paired evaluation. The two external rows are historical references rather than current state-of-the-art claims.

The three historical convolutional aggregate records report 94.0%, 94.4%, and 92.8%. Their narrow spread is descriptive bounded evidence across independently trained checkpoints.

Separately, the paired ensemble evaluation re-evaluates the three members at 93.6%, 95.0%, and 93.2%, for a 93.9% mean. Elementwise prediction averaging reaches 96.2%. This is 2.3 points above that paired mean and 1.2 points above its strongest member. The gain is largest on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one 19.2M-parameter policy.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It is an existence claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization or state of the art.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a high-order coefficient table. Each entry says how strongly one combination of query, key, and value coordinates contributes to an output. It is *weight-derived* because no examples or fitted probe are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit

numerator and denominator factors carry the normalization. The rational extension agrees to 3.61×10^{-16} . We then reconstruct all four vision and eight joint attention modules in each of three trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the worst absolute difference is 2.23×10^{-5} . This residual comes from the deployed implementation’s intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

We separately certify the three released convolutional checkpoint files named in the historical capability aggregate records in Section 3. Their vision stacks contain no attention, so we independently rebuild all eight joint modules and all twelve heads per module from the saved weights. Sixteen deterministically selected cache inputs per checkpoint yield 384 module cases and 4,608 head-input cases. Every output is finite and the worst relative ℓ_2 error is 6.36×10^{-16} , far below the frozen 10^{-6} gate. This same-checkpoint certificate covers the joint-attention equation at each normalized layer input. It does not rebuild the convolutional stem, feed-forward blocks, residual composition, final normalization, or action head as one end-to-end contraction.

Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15} values. We instead contract the CP factors into an $O(d^2)$ reduced Gram. This smaller matrix plays the role of a covariance matrix: its leading eigenvectors rank the directions that carry the most coefficient energy without constructing the full table. At toy multi-head scale, it matches brute-force materialization to 2.13×10^{-14} .

4.2 Trained policy result

The reconstruction certificate above spans three ViT checkpoints. The following subspace-ranking experiment is a descriptive analysis of the seed-0 ViT checkpoint. We apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all evaluation examples. The weights choose the subspace. Real cached activations are used only afterward to measure how faithfully that subspace preserves the layer’s computation.

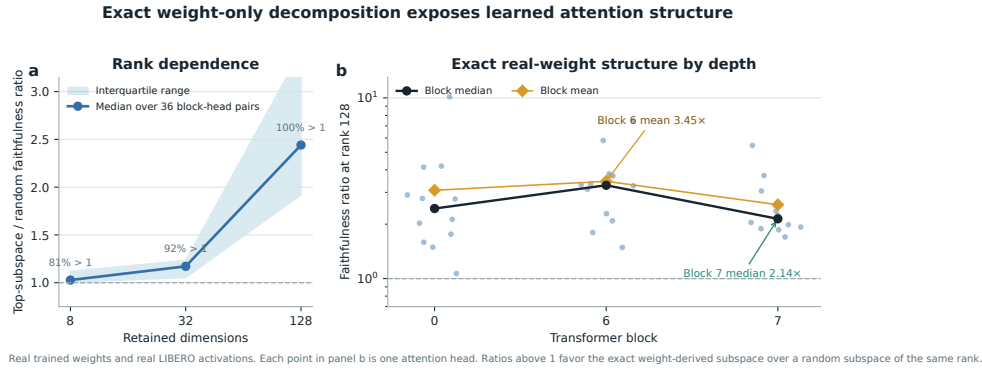


Figure 4: Exact weight-derived attention structure. Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7. The 36 block-head pairs are architectural units within one checkpoint, not independent model-level replicates.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

5 Related work

VLA capability. OpenVLA established an open pretrained VLA and standardized LIBERO comparisons [1]. OpenVLA-OFT showed that continuous action chunks and simple regression can substantially improve adaptation and control efficiency [2]. These systems use large pretrained backbones. χ -VLA instead studies a compact from-scratch specialist under an algebraic restriction.

Tensor and polynomial networks. Polynomial networks and tensor networks use explicit multilinear structure for representation and learning [12–14]. Bilinear MLP work showed that weight eigendecomposition can expose low-rank features and circuits [3]. χ -nets introduced ODT for compositional bilinear networks [4]. We extend the setting to a continuous closed-loop policy and give an exact reduced construction for individual softmax-free attention layers.

Causal mechanisms and policy analysis. Bilinear IMPALA policies have combined competitive ProcGen control with weight-derived eigenfilters, followed by activation probing and control [5]. This is the closest architectural precedent for weight-based policy analysis. Recent work steers pretrained VLAs through causally identified activation directions [6]. Sparse feature circuits identify and edit causal feature graphs in conventional language models [7], while nonlinear causal reductions recover behavioral patterns in reinforcement-learning policies from rollout perturbations [8]. These results make a universal separation from conventional policies untenable. Our intended distinction is empirical: exact weight auditing in a strong continuous vision-language policy.

VLA robustness. Recent evaluations show that standard LIBERO success can coexist with weak robustness and language use [9–11]. We therefore treat LIBERO-Object as a bounded specialist capability setting rather than evidence of broad generalization.

6 Limitations and conclusion

χ -VLA demonstrates that a complete rational tensor policy can reach strong specialist closed-loop capability. The real checkpoint passes a mechanical operator audit. Exact weight-derived analysis now crosses individual attention layers.

The scope is deliberately specific. LIBERO-Object is one specialist suite with fixed task instructions, so these results do not establish broad robotic generalization or robust instruction grounding. Exact reconstruction is certified layer by layer and does not yet yield one compact symbolic contraction of the full policy. The three-checkpoint ensemble trades three forward passes for its strongest capability result.

Within that scope, polynomial and rational restrictions are compatible with strong closed-loop control and provide a precise weight-derived analysis surface. The released convolutional files support the joint-attention equation certificate and are named in the historical capability aggregate records. Separate ViT checkpoints support the reduced-Gram subspace screen. We do not claim a direct coefficient-edit interface.

Reproducibility. Training and evaluation use fixed task lists, explicit camera and action conventions, canonical-state manifests, checkpointed moving-average weights, and per-run result files. Historical Modal cache-backed analyses read dataset-native task metadata. The replacement Athena pipeline pins that metadata and maps dataset identifiers to official identifiers before defining task splits. A source-level audit compared all 271,996 cached frames across Object, Spatial, Goal, and LIBERO-10 with their pinned LeRobot revisions. Task identifiers, language joins, states, actions, and resized images matched exactly for every frame. The anonymous artifact releases raw structural-certificate JSONs, immutable SHA-256 identities, and a standard-library one-command verifier. Preflight checks fail if canonical initial states cannot be loaded.

References

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A Supporting results and decisions

A.1 Per-task ensemble behavior

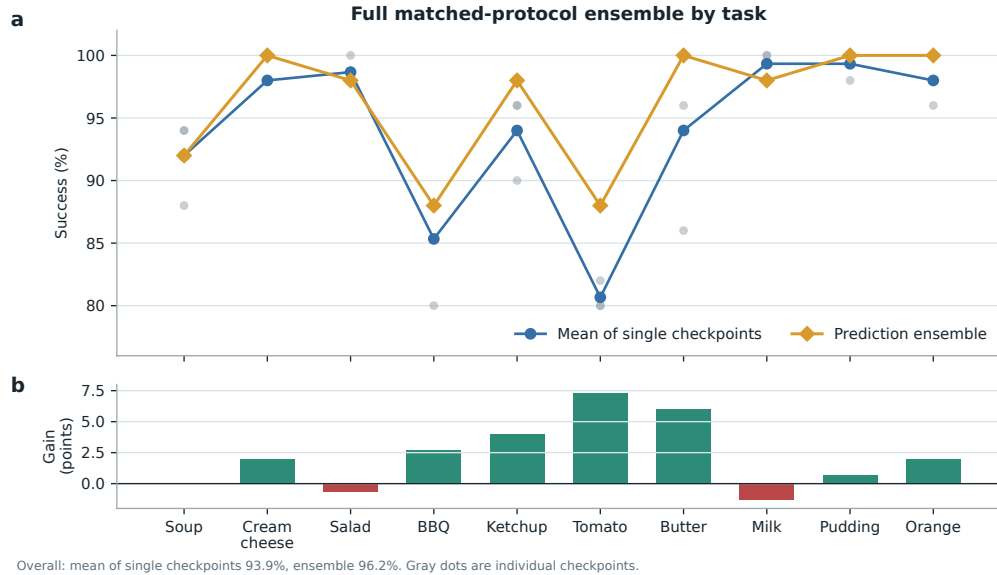


Figure 5: **Per-task effect of prediction averaging in the paired ensemble evaluation.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 19.2M-parameter policy.

253 **A.2 Expanded attention screen**

254 Within the same seed-0 ViT checkpoint, the corrected fixed-control attention analysis covers all
255 eight blocks and twelve heads. At rank 128, 83/96 block-head pairs favor the exact weight-derived
256 subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth
257 controls. At ranks 8 and 32, only 54/96 and 60/96 pairs favor the exact subspace. The effect is
258 therefore depth-dependent and strongest at the wider tested rank.

259 A.3 Numerical scope of rational conversion

260 The deployed degree- $[2/2]$ least-squares rational fit is exact as a rational computation even though it
 261 approximates RMS normalization. Its numerator has coefficients (5.32997, 7.53552, 0.327274) and
 262 the denominator has coefficients (1, 9.64827, 2.60664). Every denominator coefficient is positive,
 263 which certifies $Q(v) > 0$ for all $v \geq 0$. We instrument all 53 RationalNorm module instances,
 264 each invoked once per forward pass, on 4,096 deterministic inputs for each of three convolutional
 265 checkpoints. Across 544,542,720 activation rows, the minimum observed denominator is 1.0016 and
 266 no row is nonfinite. Comparing deployed float32 normalization with an otherwise identical pass that
 267 recomputes only RationalNorm scales in float64, the largest checkpoint-level primary metric, defined
 268 as the maximum of normalized- and physical-action NRMSE, is 2.414×10^{-6} against a frozen 10^{-3}
 269 gate.

270 The certificate also bounds what the operator approximates. Observed normalized mean-square ratios
 271 v span 1.66×10^{-4} to 17.17, outside the nominal fit interval $[0.1, 10]$. Across site-checkpoint pairs,
 272 the minimum coverages are 60.018% in that interval and 60.022% with rational-versus-exact-RMS
 273 scale error at most 3.4%. The frozen 99% secondary coverage gates therefore do not pass. The
 274 positive claim is therefore global absence of nonnegative denominator poles and cache-conditioned
 275 floating-point fidelity to the deployed rational operator, not uniform agreement with exact RMS
 276 scaling. Matched alternative-normalizer runtime overhead remains unmeasured.

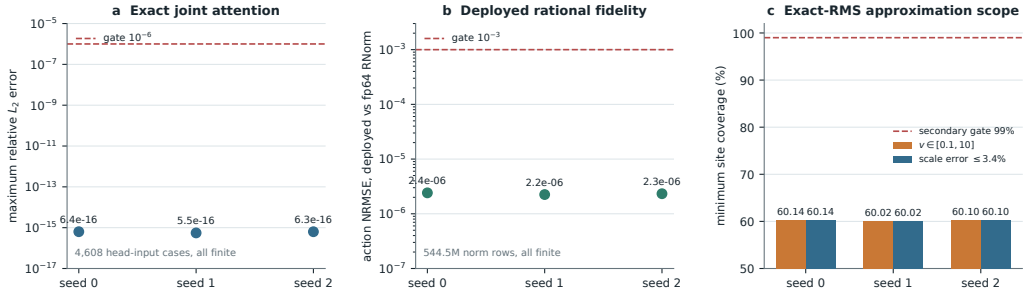


Figure 6: **Frozen structural certificates on the released convolutional files named in the historical capability aggregate records.** Panels (a) and (b) pass their primary gates with large margins. Panel (a) reconstructs all 4,608 audited joint-attention head-input cases. Panel (b) compares deployed float32 normalization with an otherwise identical pass that recomputes only RationalNorm scales in float64. Panel (c) reports the separately frozen exact-RMS approximation coverage, which does not pass its 99% secondary gate.